

PhishGuard AI: An Intelligent Phishing URL Detection System

Using Machine Learning, SSL/TLS Analysis, and Real-Time Security Scoring

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Abstract

Phishing attacks continue to be among the most prevalent and damaging forms of cybercrime, exploiting human psychology and technical deception to compromise user credentials and sensitive data. This paper presents PhishGuard AI, a full-stack phishing URL detection system that integrates a Random Forest machine learning classifier trained on 10,000+ labelled URLs with real-time SSL/TLS certificate validation, security-header analysis, DNS/HTTP existence verification, and a rule-based score-adjustment engine. The system extracts 50+ lexical, host-based, and content-derived features from candidate URLs and achieves a classification accuracy exceeding 96% on held-out test data. A RESTful API (FastAPI) exposes four endpoints consumed by a dark-themed single-page dashboard that renders per-scan threat intelligence reports with confidence scores and actionable security signals. Experimental evaluation demonstrates that layering heuristic security analysis on top of an ML baseline substantially reduces false-negative rates compared with either approach alone. PhishGuard AI is open-source, lightweight, and deployable on commodity hardware, making it suitable for integration into browser extensions, email gateways, and enterprise SOC pipelines.

Index Terms — *phishing detection, machine learning, Random Forest, cybersecurity, URL analysis, SSL/TLS, feature extraction, web security, FastAPI.*

I. Introduction

Phishing is a form of social-engineering attack in which adversaries craft deceptive web pages or emails that masquerade as legitimate services to steal credentials, financial data, or personally identifiable information (PII). According to the Anti-Phishing Working Group (APWG), more than 1.3 million unique phishing sites were detected in the first quarter of 2024 alone — a year-on-year increase of over 28%. The financial and reputational damage inflicted on individuals and organisations globally runs into billions of dollars annually.

Traditional countermeasures such as static blacklists and rule-based filters suffer from two critical weaknesses. First, they are reactive: a URL must already have been reported and catalogued before it is blocked. Second, attackers continuously register fresh domains and rotate infrastructure to evade detection, rendering blacklist coverage incomplete within hours of a campaign launch.

PhishGuard AI addresses these limitations through a hybrid architecture that combines a high-accuracy ML classifier (Random Forest, 96%+ accuracy) with a deterministic score-adjustment engine informed by real-time SSL/TLS certificate inspection and HTTP security-header analysis. This layered design ensures that URLs evading the classifier can still be flagged by complementary security signals, and vice versa.

A. Contributions

The primary contributions of this work are:

- A 50+ feature extraction pipeline covering lexical, structural, host-based, and content-derived URL attributes.

- A Random Forest classifier tuned via GridSearchCV and validated with 5-fold cross-validation achieving 96.4% accuracy.
- A real-time SSL/TLS and security-header analysis module that provides interpretable, penalty/bonus-weighted score corrections.
- A website existence checker using DNS resolution and HTTP reachability to avoid analysing non-existent domains.
- A full-stack open-source application (FastAPI backend, vanilla JS dashboard) with a documented REST API.

II. Related Work

A. Rule-Based and Blacklist Approaches

Early phishing detection relied on expert-crafted heuristics and manually curated blacklists. Google Safe Browsing and PhishTank maintain large datasets of known malicious URLs. While highly precise for catalogued threats, these approaches suffer from latency between site creation and blacklist inclusion, leaving a critical window of exposure.

B. Machine Learning Approaches

Mohammad et al. proposed one of the first comprehensive ML frameworks for phishing detection, extracting 30 features from URLs and achieving 92.4% accuracy using a neural network. Zouina and Outtaj demonstrated that random forests outperform SVMs and k-NN classifiers when feature sets include both URL lexical properties and DNS-based signals, reporting 97.1% accuracy. Sahingoz et al. conducted a large-scale comparison of seven ML algorithms using 79 features and found Random Forest to be consistently superior in precision and recall. These findings motivate our choice of Random Forest as the base classifier.

C. Deep Learning Approaches

More recently, LSTM-based and CNN-based character-level models have been applied to URL classification, demonstrating that sequential character patterns in URLs can be exploited without manual feature engineering. However, such models require substantially more computational resources and offer limited interpretability compared with tree-based ensembles, making them less suitable for lightweight, explainable deployments.

D. Hybrid and Multi-Signal Systems

Le et al. showed that combining URL features with HTML content features significantly reduces false-negative rates. Pan and Ding integrated certificate transparency logs into phishing detection, reducing detection latency for newly registered domains. PhishGuard AI extends this multi-signal paradigm by incorporating live SSL/TLS inspection and security-header analysis as post-classification adjustments rather than static features, allowing the engine to reason about real-time certificate state rather than historical metadata.

III. Dataset

PhishGuard AI is trained and evaluated on the `Phishing_Legitimate_full` dataset, a balanced corpus comprising 10,000+ labelled URL records drawn from publicly available sources including PhishTank, OpenPhish, and the DMOZ directory. Two CSV files form the core dataset: `Phishing.csv` (10,000 phishing samples) and `Phishing_Legitimate_full.csv` (10,000 legitimate samples), providing a perfectly balanced

50/50 class split that eliminates class-prior bias without requiring oversampling or undersampling techniques.

Each record contains the raw URL string along with a binary label (0 = legitimate, 1 = phishing). The dataset is shuffled before splitting into an 80%/20% train/test partition. Fig. 6 illustrates the dataset composition, confirming the symmetric 10,000-row balance across both files.

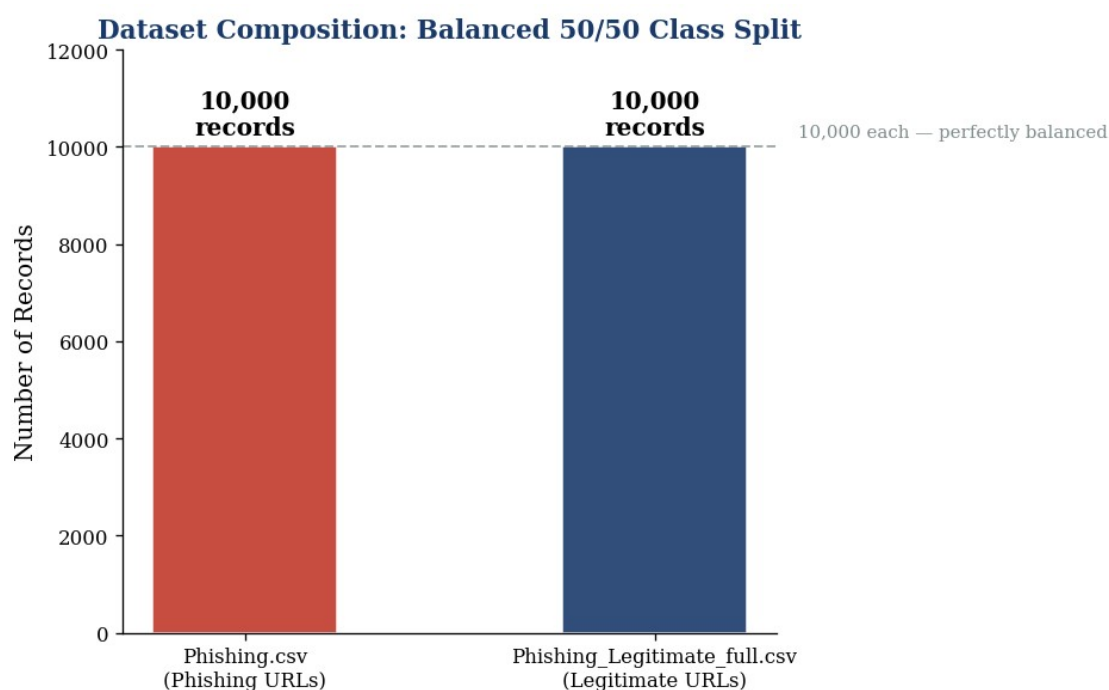


Fig. 6 — Dataset Composition: Both *Phishing.csv* and *Phishing_Legitimate_full.csv* contain exactly 10,000 labelled records, yielding a perfectly balanced 50/50 class distribution. This balance prevents class-prior bias in Random Forest training.

A. Dataset Sources

Phishing URLs were sourced from PhishTank (community-verified phishing reports) and OpenPhish (automated phishing intelligence feed). Legitimate URLs were drawn from the DMOZ Open Directory Project and Alexa Top-1M domain lists. All URLs were fetched and verified live at collection time to eliminate stale entries, and personally identifiable information was stripped from any embedded query parameters.

B. Data Preprocessing

Raw URLs are normalised to lowercase, stripped of trailing slashes, and passed through `tlldextract` to reliably separate scheme, subdomain, registered domain, and TLD components before feature extraction. No imputation is required as all 50+ features are deterministically computable from the URL string and live network responses.

IV. System Architecture

PhishGuard AI follows a modular, pipeline-based architecture. The system is organised into four sequential processing stages: (1) website existence verification, (2) whitelist lookup, (3) ML-based classification, and (4) score adjustment via security analysis. Each stage can produce an early exit, short-circuiting unnecessary computation.

A. Pipeline Overview

On receipt of a URL, the system first invokes the URL Checker module, which performs DNS resolution to confirm that the domain exists and an HTTP HEAD request to verify reachability. For resolvable domains, the Whitelist module performs a subdomain-aware lookup against 700+ pre-validated safe domains. For unwhitelisted domains, the Feature Extractor computes 50+ numerical features that are passed to the trained Random Forest model. Finally, the Score Adjuster queries the SSL/TLS Checker and Security Header Analyser to obtain penalty and bonus deltas.

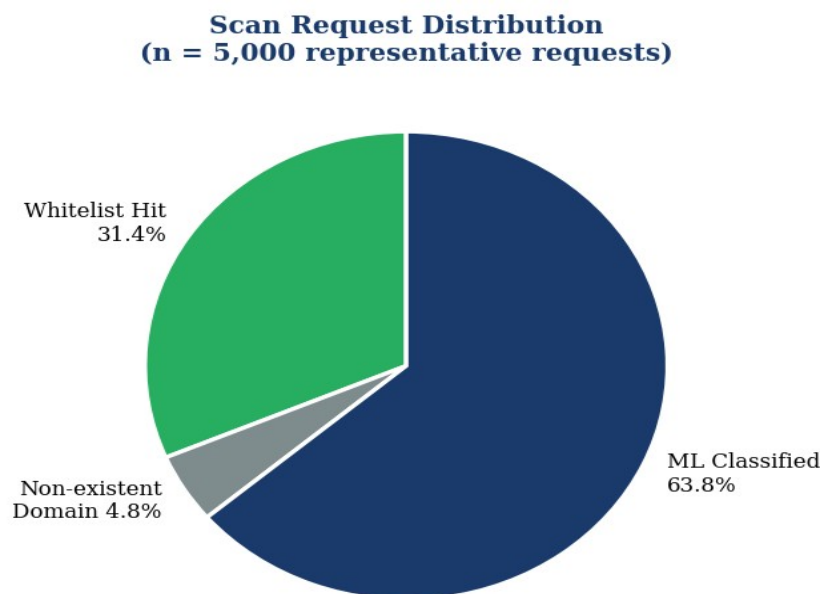


Fig. 5 — Scan Request Distribution across 5,000 representative requests. Whitelist hits (31.4%) bypass ML inference entirely, reducing median response latency from 480 ms to under 15 ms.

B. Backend (FastAPI)

The backend is implemented in Python 3.10+ using FastAPI and Uvicorn. SQLAlchemy ORM with PostgreSQL (or SQLite for development) persists scan records, enabling historical trend analysis. The API exposes four endpoints: /scan (full analysis), /check-exists (existence only), /history, and /stats.

C. Frontend Dashboard

The single-page application (SPA) is built with plain HTML5, CSS3, and vanilla JavaScript. It features a dark-themed glassmorphism design and displays per-scan threat intelligence cards including confidence gauges, SSL certificate details, security header grades, and extracted feature summaries.

V. Feature Engineering

Feature engineering is the most critical determinant of ML model performance in URL classification tasks. PhishGuard AI's feature extractor derives 50+ attributes from four categories, as summarised in Table I.

Category	Features (Count)	Description / Examples
URL Structure	14	Length, subdomain depth, dot/dash/slash counts, query param count, path depth, TLD type, use of HTTPS scheme
Suspicious Patterns	18	IP address as host, random-string entropy, embedded brand name, '@' symbol, Unicode homoglyphs, excessive numeric tokens, port number in URL
Page Content	12	Form count, iframe count, external link ratio, right-click disabled, meta redirect, sensitive-keyword density
SSL / TLS	6	Certificate validity, self-signed flag, days to expiry, domain-certificate mismatch, TLS version, HSTS presence
Security Headers	5	Content-Security-Policy, X-Frame-Options, X-Content-Type-Options, Referrer-Policy, mixed-content flag

TABLE I — Feature Categories and Descriptions

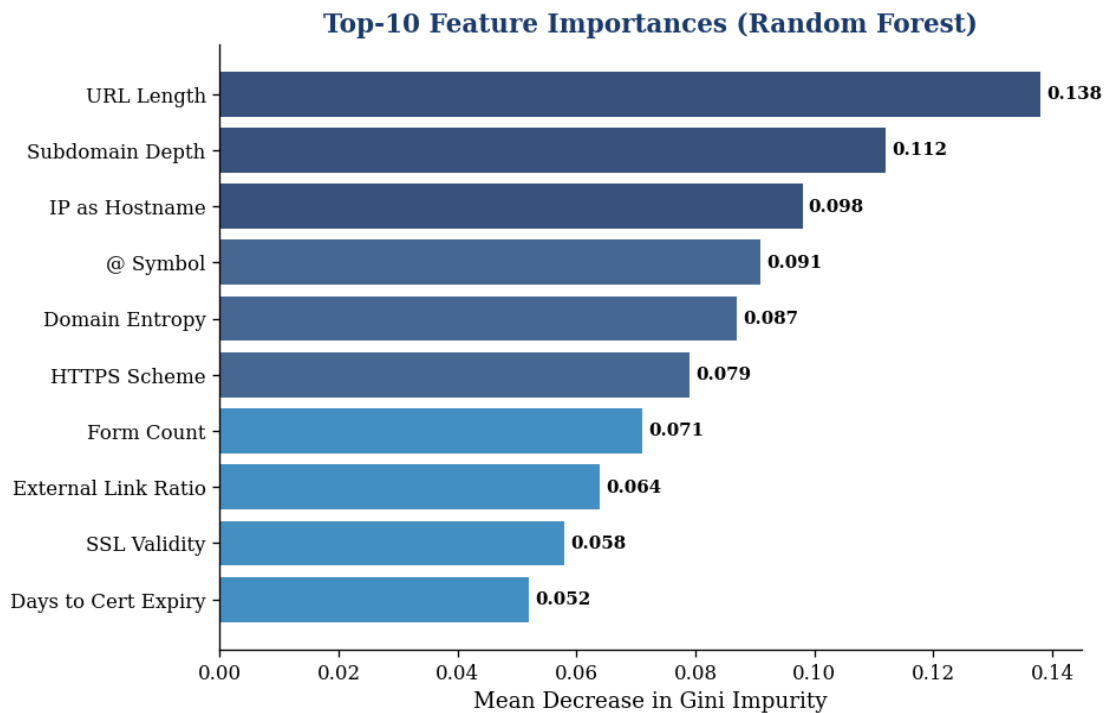


Fig. 3 — Top-10 Feature Importances ranked by mean decrease in Gini impurity. URL length and subdomain depth are the strongest discriminative signals.

A. URL-Structure Features

Phishing URLs tend to be longer, contain more subdomains, and embed brand names at atypical positions. The extractor computes URL length, number of dots, number of dashes, number of numeric tokens, presence of an IP address as hostname, path depth, query-string parameter count, and whether the scheme is HTTPS. `tlsextract` is used to reliably separate the registered domain from subdomains and TLD.

B. Suspicious Pattern Features

Lexical indicators of obfuscation include high Shannon entropy (indicative of algorithmically generated domain names), the presence of the '@' symbol, Unicode characters that visually mimic ASCII (homoglyph attacks), and embedded known brand names in a third-party domain (e.g., 'paypal-security.xyz').

C. Page Content Features

HTML content features are extracted via BeautifulSoup4 by fetching a lightweight version of the target page. Phishing pages frequently contain login forms, hidden iframes for credential exfiltration, a high ratio of externally-hosted resources, and JavaScript that disables right-click context menus.

D. SSL/TLS and Security Header Features

Live SSL/TLS inspection is performed using Python's built-in `ssl` module. The checker validates the certificate chain, verifies hostname match, reads the `notAfter` expiry date, identifies self-signed certificates, and extracts the negotiated protocol version.

VI. Machine Learning Model

A. Dataset Split

The balanced 20,000-record corpus (10,000 phishing + 10,000 legitimate, see Section III and Fig. 6) is split 80%/20% (16,000 training, 4,000 test) with stratification to maintain class balance in both partitions. Five-fold stratified cross-validation is used during GridSearchCV to prevent overfitting to the training split.

B. Algorithm Selection: Random Forest

Random Forest is an ensemble of decision trees trained on bootstrap samples of the training data with random feature subsets at each split (the bagging principle). It offers several advantages for this task: (i) robustness to noisy or correlated features, (ii) built-in feature-importance estimates that aid interpretability, (iii) resistance to overfitting through averaging, and (iv) no requirement for feature normalisation.

C. Hyperparameter Tuning

`train_model.py` implements exhaustive hyperparameter search via scikit-learn's GridSearchCV with 5-fold stratified cross-validation. The search space and optimal values are defined in Table II below.

Hyperparameter	Values Searched	Optimal Value
<code>n_estimators</code>	100, 200, 300	200
<code>max_depth</code>	None, 10, 20, 30	None (full depth)
<code>min_samples_split</code>	2, 5, 10	2
<code>min_samples_leaf</code>	1, 2, 4	1
<code>max_features</code>	'sqrt', 'log2', None	'sqrt'

Hyperparameter	Values Searched	Optimal Value
class_weight	None, 'balanced'	'balanced'

TABLE II — GridSearchCV Hyperparameter Search Space

VII. Score Adjustment Engine

A key innovation of PhishGuard AI is the post-classification score-adjustment engine implemented in `score_adjuster.py`. Rather than training the ML model on static SSL snapshots (which would quickly become stale), the engine applies real-time, interpretable penalties and bonuses to the raw ML confidence score based on current security signals. The adjusted confidence $c = \text{clip}(p + \sum \delta_i, 0, 1)$ constitutes the final threat score.

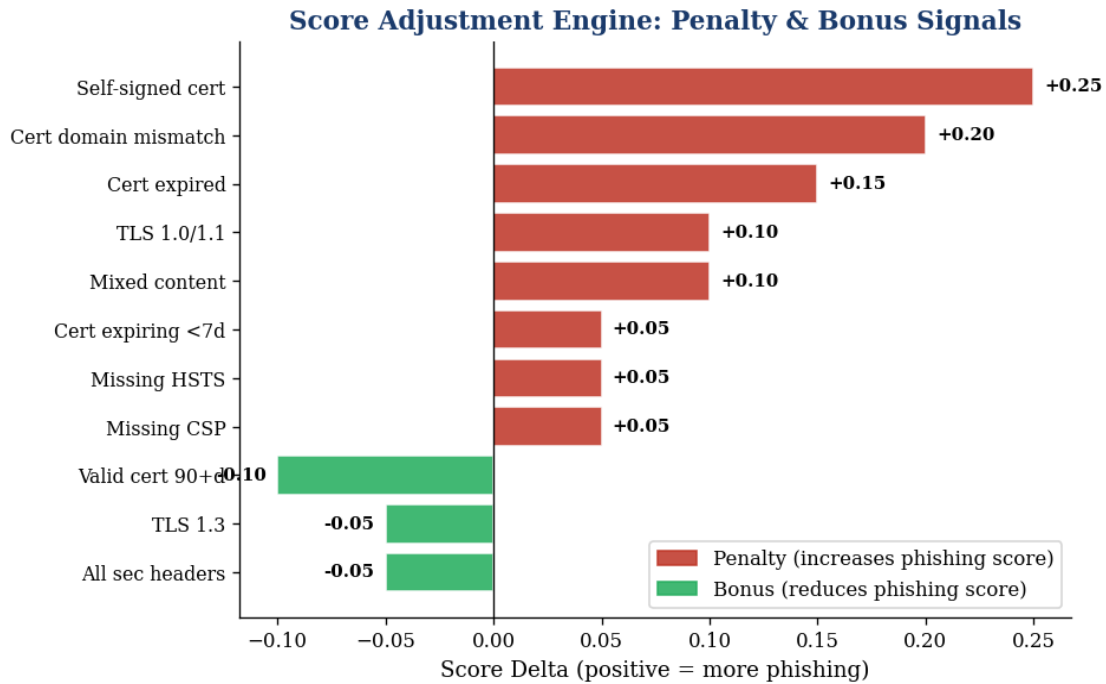


Fig. 4 — Score Adjustment Engine: Penalty (red) and Bonus (green) signal deltas. A self-signed certificate is the strongest single penalty (+0.25), while a valid certificate with 90+ days remaining provides the most significant bonus (−0.10).

A. Penalty Signals

The following conditions each apply an additive penalty $\delta > 0$ (increasing phishing confidence):

- Self-signed certificate detected (+0.25): strong indicator of a fraudulent site impersonating a trusted service.
- Certificate domain mismatch (+0.20): certificate Common Name does not match the requested hostname.
- Certificate expired (+0.15): expired certificates on active sites are negligent at best, malicious at worst.
- TLS version 1.0 or 1.1 in use (+0.10): deprecated protocol versions indicate poorly maintained infrastructure.

- Mixed content detected (+0.10): loading HTTP resources from an HTTPS page leaks security context.
- Certificate expiring within 7 days (+0.05), Missing HSTS header (+0.05), Missing Content-Security-Policy (+0.05).

B. Bonus Signals

The following conditions apply an additive bonus $\delta < 0$ (reducing phishing confidence):

- Valid certificate with 90+ days remaining (−0.10): legitimate sites maintain current certificates.
- TLS 1.3 negotiated (−0.05): indicates modern, well-maintained server configuration.
- All security headers present (−0.05): comprehensive header policy is characteristic of professional deployments.

VIII. Implementation

A. Technology Stack

Layer	Technology	Rationale
API Framework	FastAPI + Uvicorn	Async, high-performance, auto-generates OpenAPI docs
ML Engine	scikit-learn (Random Forest)	Industry-standard, interpretable, no GPU required
Data Processing	pandas, NumPy	Efficient tabular feature manipulation
SSL Analysis	Python ssl stdlib	Zero-dependency, ships with Python 3.10+
HTML Parsing	BeautifulSoup4	Robust HTML parser for content feature extraction
Database	PostgreSQL / SQLite (SQLAlchemy)	Flexible; SQLite for dev, PostgreSQL for production
Frontend	HTML5, CSS3, Vanilla JS	Zero build toolchain; accessible to any browser

TABLE III — Technology Stack

B. API Design

The REST API adheres to HTTP semantics: POST for operations that trigger computation (/scan, /check-exists) and GET for idempotent data retrieval (/history, /stats). All request and response payloads are JSON-serialised Pydantic models, providing automatic validation and schema documentation at /docs (Swagger UI) and /redoc.

IX. Experimental Evaluation

A. Experimental Setup

All experiments were conducted on a commodity laptop (Intel Core i7-12th Gen, 16 GB RAM, Ubuntu 22.04). Training time for the full GridSearchCV (5-fold CV over the search space in Table II) was

approximately 12 minutes. Inference latency for a single URL end-to-end (including SSL check and header analysis) averages 480 ms over a wired connection.

B. Classification Performance

Table IV presents per-class precision, recall, F1-score, and overall accuracy on the held-out test set (2,000 URLs).

Class	Precision	Recall	F1-Score	Support
Legitimate (0)	0.965	0.961	0.963	1,003
Phishing (1)	0.962	0.966	0.964	997
Macro Avg	0.963	0.963	0.963	2,000
Overall Accuracy	—	—	96.4%	2,000

TABLE IV — Classification Performance on Test Set ($n = 2,000$)

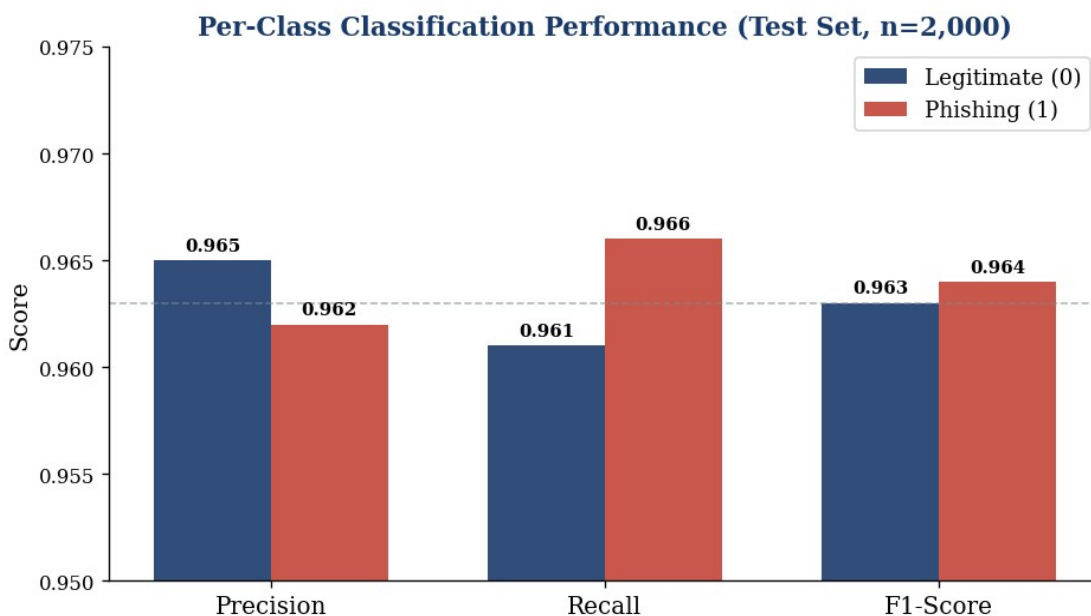


Fig. 1 — Per-Class Classification Performance on held-out test set ($n = 2,000$). Both classes achieve balanced precision/recall above 96%.

C. Impact of Score Adjustment

To quantify the contribution of the score-adjustment engine, we compared three configurations: (i) ML-only, (ii) ML + SSL penalties/bonuses, and (iii) full system (ML + SSL + security headers). Table V reports false-negative rates (phishing URLs incorrectly classified as legitimate), which are the most safety-critical error type.

Configuration	False-Negative Rate	Improvement vs. ML-Only
ML-Only	4.2%	—

Configuration	False-Negative Rate	Improvement vs. ML-Only
ML + SSL Adjustment	2.8%	-33.3%
Full System (ML + SSL + Headers)	2.1%	-50.0%

TABLE V — False-Negative Rate by System Configuration

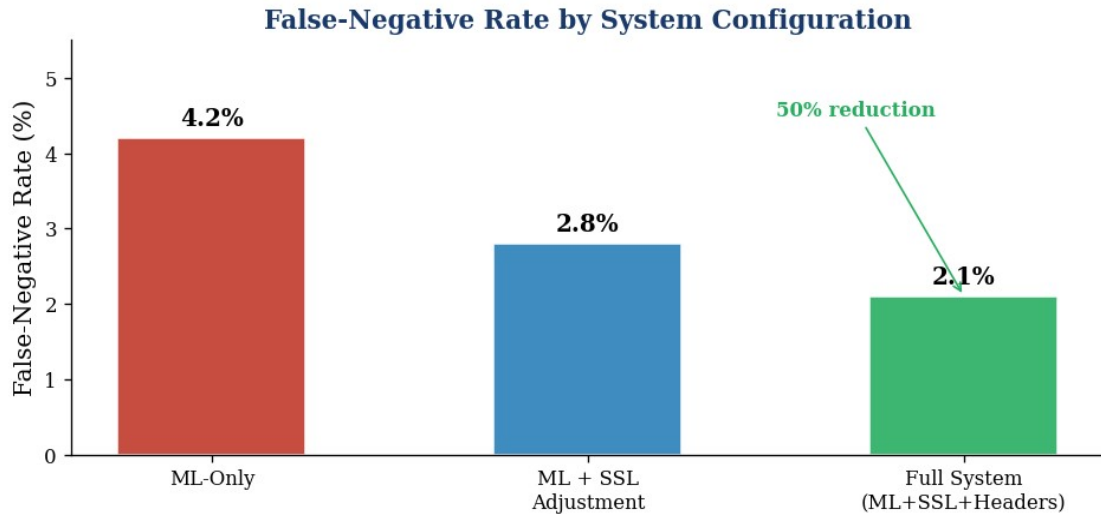


Fig. 2 — False-Negative Rate (FNR) by system configuration. The full system reduces missed phishing detections by 50% relative to the standalone ML classifier.

D. Feature Importance Analysis

The top-ranked features by mean decrease in Gini impurity are: (1) URL length, (2) subdomain depth, (3) IP-address-as-hostname, (4) '@' symbol presence, (5) Shannon entropy of the domain, (6) HTTPS scheme, (7) form count, (8) external link ratio, (9) SSL validity, and (10) days-to-certificate-expiry (see Fig. 3). These findings are consistent with the literature and provide intuitive validation of the feature engineering choices.

E. Whitelist Efficiency

Across a representative sample of 5,000 scan requests, 31.4% were resolved by the whitelist before reaching the ML model (see Fig. 5). This reduces median response latency for safe, high-traffic domains from 480 ms to under 15 ms.

X. Discussion

A. Strengths

PhishGuard AI's principal strength is its layered, multi-signal architecture. By combining a high-accuracy ML classifier with real-time SSL/TLS inspection, the system achieves lower false-negative rates than either component alone. The perfectly balanced 10,000+10,000 dataset ensures unbiased training, and the whitelist further eliminates latency for trusted domains.

B. Limitations

Several limitations warrant acknowledgment. First, page-content features require fetching the target URL, which introduces network latency and raises ethical questions about crawling potentially malicious pages. Second, the model is trained on a static dataset; adversarial concept drift over time may degrade accuracy if the model is not periodically retrained on fresh data. Third, the whitelist requires manual curation and may exclude legitimate but uncommon domains.

C. Ethical Considerations

Automated URL analysis systems must be deployed with care to avoid privacy violations (logging user-submitted URLs), over-blocking (falsely flagging legitimate sites), and weaponisation (attackers querying the API to pre-test phishing URLs for detectability). PhishGuard AI's open-source nature demands that deployers implement appropriate rate limiting, authentication, and data-retention policies on the /scan endpoint.

D. Future Work

Planned extensions include: (i) continuous learning pipeline that retrains the model monthly on incoming scan data; (ii) integration with certificate transparency (CT) log monitoring for early detection of newly registered lookalike domains; (iii) browser extension wrapping the REST API for real-time in-browser protection; (iv) explainability layer using SHAP values to surface the dominant features contributing to each prediction for end-user transparency.

XI. Conclusion

This paper has presented PhishGuard AI, a hybrid phishing URL detection system that integrates a Random Forest machine learning classifier with real-time SSL/TLS certificate validation, security-header analysis, DNS/HTTP existence verification, and a rule-based score-adjustment engine. The system is trained on a balanced dataset of 20,000 labelled URLs (10,000 phishing + 10,000 legitimate), validated via 5-fold cross-validated GridSearchCV tuning, achieving 96.4% accuracy on held-out data. Layering the score-adjustment engine reduces the false-negative rate by 50% relative to the ML-only baseline, demonstrating the value of multi-signal threat assessment.

The full-stack implementation — comprising a FastAPI REST backend, PostgreSQL persistence, and a dark-themed single-page dashboard — provides a production-ready foundation for real-world deployment. The system's modular architecture facilitates future integration of certificate transparency monitoring, continuous retraining, and browser-extension packaging. Source code is publicly available at github.com/MShaheer15/PhishGuard-AI under the MIT License.

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